LAB EXPERIMENT RECORD

Set 29: Student Academic Performance

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424049
Date	22/10/2026
Day	Thursday

AIM

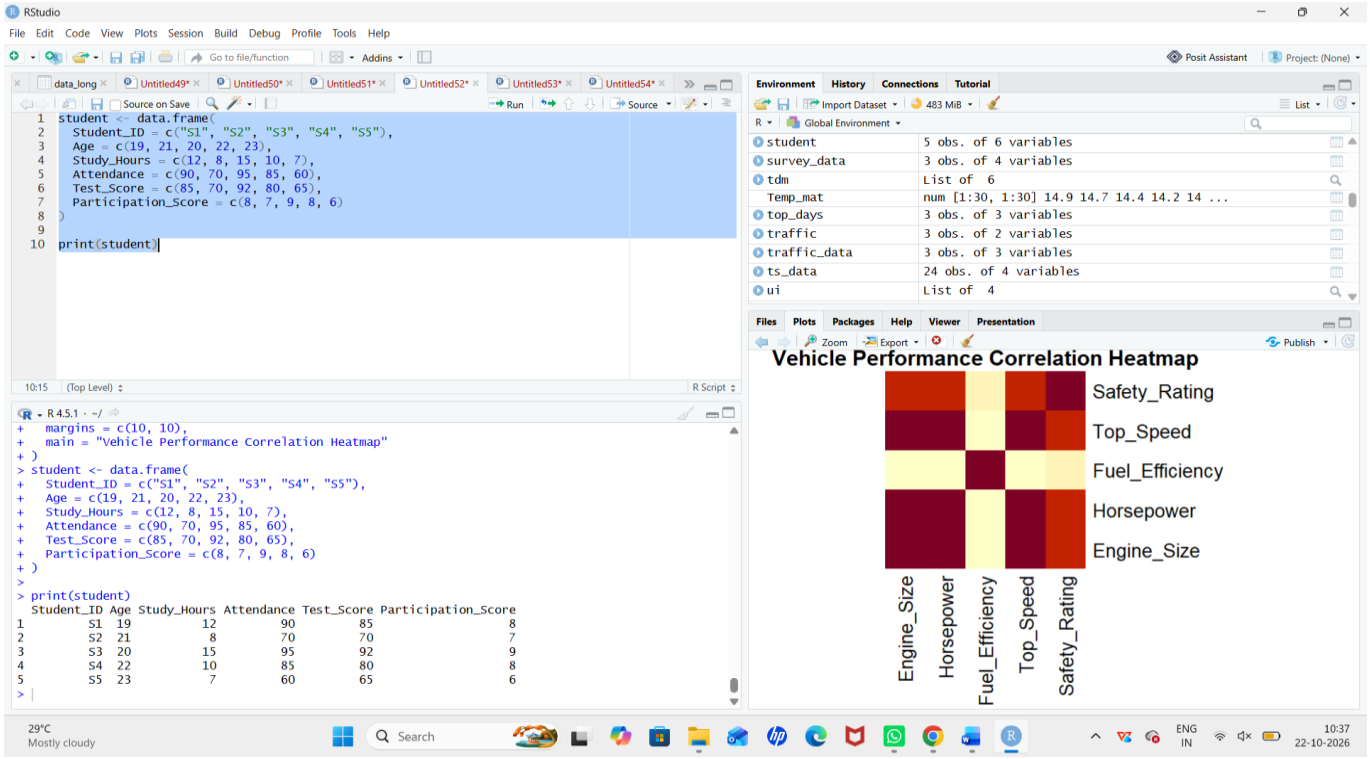
To analyze student academic performance using R and Tableau by creating a stacked area chart, boxplot, density plot, and interactive scatterplot to understand the relationships between study hours, attendance, test scores, and participation scores

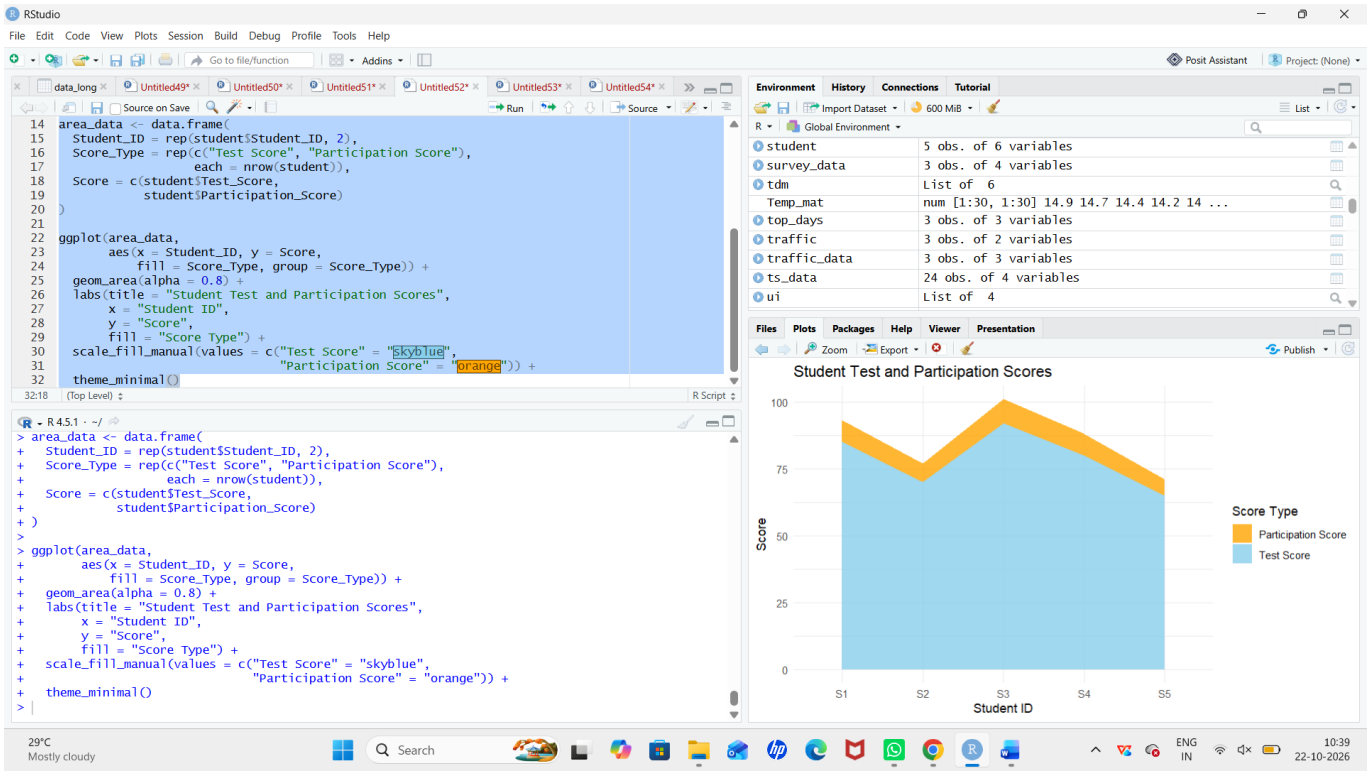
1.

```
student <- data.frame(
  Student_ID = c("S1", "S2", "S3", "S4", "S5"),
  Age = c(19, 21, 20, 22, 23),
  Study_Hours = c(12, 8, 15, 10, 7),
  Attendance = c(90, 70, 95, 85, 60),
  Test_Score = c(85, 70, 92, 80, 65),
  Participation_Score = c(8, 7, 9, 8, 6)
)

print(student)
library(ggplot2)

area_data <- data.frame(
  Student_ID = rep(student$Student_ID, 2),
```

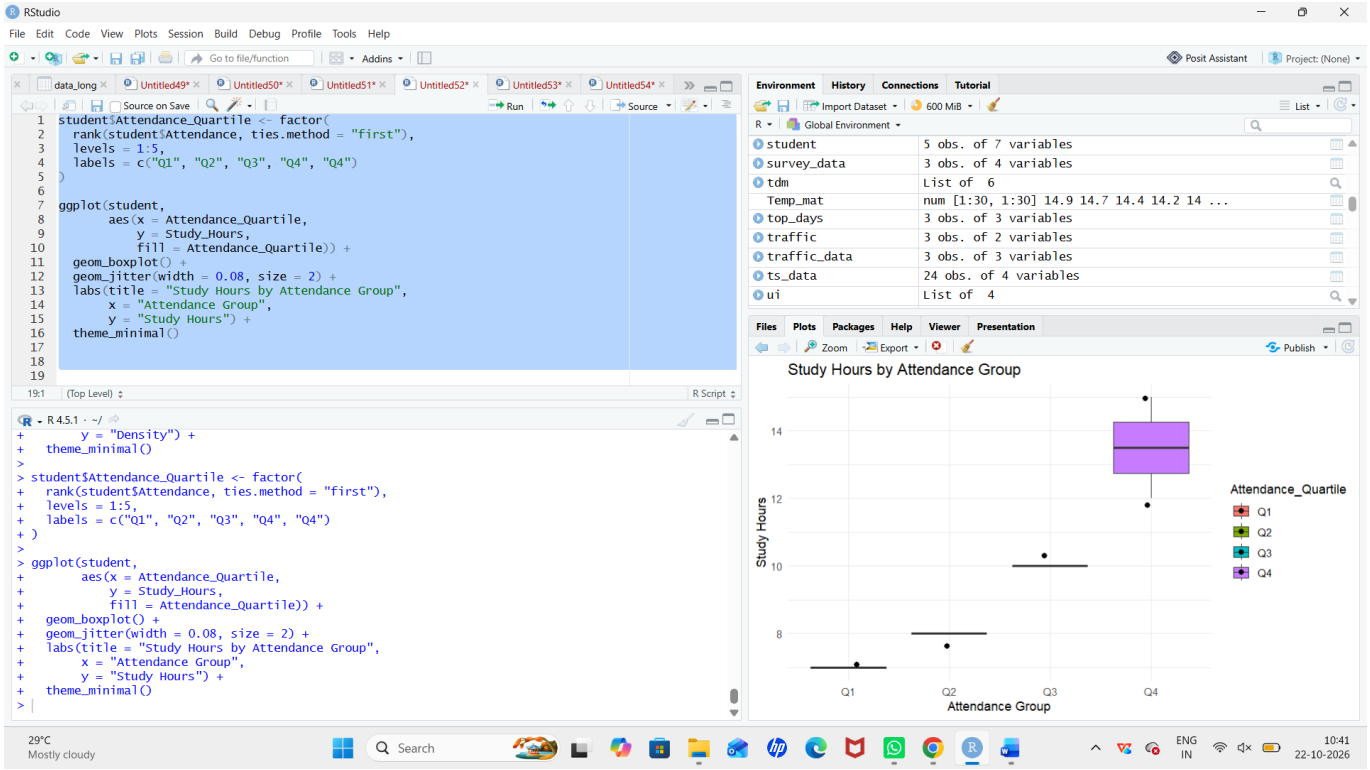



RStudio
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area_data <- data.frame(
Student_ID = rep(student$Student_ID, 2),
Score_Type = rep(c("Test Score", "Participation Score"),
each = nrow(student)),
Score = c(student$Test_Score,
student$Participation_Score)
)
ggplot(area_data,
aes(x = Student_ID, y = Score,
fill = Score_Type, group = Score_Type)) +
geom_area(alpha = 0.8) +
labs(title = "Student Test and Participation Scores",
x = "Student ID",
y = "Score",
fill = "Score Type") +
scale_fill_manual(values = c("Test Score" = "skyblue",
"Participation Score" = "orange")) +
theme_minimal()
Environment History Connections Tutorial
student 5 obs. of 6 variables
survey_data 3 obs. of 4 variables
tdm List of 6
Temp_mat num [1:30, 1:30] 14.9 14.7 14.4 14.2 14 ...
top_days 3 obs. of 3 variables
traffic 3 obs. of 2 variables
traffic_data 3 obs. of 3 variables
ts_data 24 obs. of 4 variables
ui List of 4
Files Plots Packages Help Viewer Presentation
Student Test and Participation Scores
Score
Student ID
Score Type
Participation Score
Test Score
29°C
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2.

```
library(ggplot2)

ggplot(student, aes(x = Test_Score)) +
  geom_density(fill = "lightblue",
               color = "blue",
               alpha = 0.7,
               adjust = 1.5) +
  geom_rug() +
  labs(title = "Density Plot of Student Test Scores",
       x = "Test Score",
       y = "Density") +
  theme_minimal()
```



3.

```
library(ggplot2)

student$Attendance_Quartile <- cut(
  student$Attendance,
  breaks = quantile(student$Attendance,
                    probs = seq(0, 1, 0.25),
                    na.rm = TRUE),
  include.lowest = TRUE,
  labels = c("Q1", "Q2", "Q3", "Q4")
)

ggplot(student,
       aes(x = Attendance_Quartile,
           y = Study_Hours,
           fill = Attendance_Quartile)) +
  geom_boxplot() +
  geom_jitter(width = 0.08, size = 2) +
  scale_fill_manual(values = c(
    "Q1" = "lightblue",
    "Q2" = "lightgreen",
    "Q3" = "orange",
    "Q4" = "pink"
  )) +
  labs(title = "Study Hours by Attendance Quartile",
       x = "Attendance Quartile",
       y = "Study Hours") +
  theme_minimal()
```

3. TABLEAU: INTERACTIVE SCATTERPLOT

Steps:

1. Open Tableau Desktop or Tableau Public.
2. Create a CSV file using the student dataset and connect to it.
3. Open a new worksheet.
4. Drag Study_Hours to Columns (X-axis).
5. Drag Test_Score to Rows (Y-axis).
6. Select Circle in the Marks dropdown.
7. Drag Student_ID to Detail.
8. Drag Participation_Score to Color and choose a color gradient.
9. Drag Age to the Filters shelf, select the required ages, and click Show Filter.
10. Add a suitable title: Student Academic Performance Dashboard.

Interpretation:

- S3 (15 study hours, score 92) performs best in this dataset.
- S5 (7 study hours, score 65) has the lowest test score.
- The scatterplot suggests a positive relationship between study hours and test scores.
- The Age filter allows users to compare students in different age groups.

Because the dataset is small, these observations do not establish causation.

PROCEDURE

1. Create the student dataset in R.
2. Generate a stacked area chart to compare test and participation scores.
3. Group attendance values and create a boxplot of study hours.
4. Create a density plot to examine the distribution of test scores.
5. Import the dataset into Tableau and create a scatterplot of study hours versus test scores.
6. Color the points by participation score, add an Age filter, and analyze the results.

RESULT

Student academic performance was analyzed using R and Tableau. The charts showed that S3 had the highest test score, participation score, study hours, and attendance, while S5 had the lowest test score and attendance. The scatterplot suggested a positive relationship between study hours and test scores. The Tableau visualization provided an interactive way to compare students by age and participation score. Since the dataset contains only five students, the findings are descriptive and should not be generalized to a larger student population.